

PhD Project Description

Title: Optimized IoRT network for enhanced data quality of IoRT cereals production practices

Project Summary: Wireless communication systems are being used in many applications, as they offer a flexible solution for maintaining connectivity between connected devices. This PhD focus on the use of wireless communications in agricultural mobile robots in Precision Farming. In such applications, robots need to be supervised during the missions on the field, and in some cases, they also need to be remotely controlled by a human operator. These robots continuously collect data and send it to a human operator who ensures that robots execute the right tasks based on the collected data.

Research Challenges: The operator's decisions depend on the quality of data collected by the robots and from other sensors deployed on the field. Even though there has been a lot of progress in telecommunication systems in the past 10 years, wireless communications are still prone to performance degradation depending on the current network conditions. For example, the presence of obstacles or high levels of interference might render the communication link unstable, which has a severe impact on the quality of experience of the operator.

Objectives: This PhD will tackle the issue of real-time monitoring of the state of the network to explain the quality of data the operator receives. The work will focus on prediction algorithms, based on machine learning, that allow the system to anticipate performance degradation and inform the operator in real-time. Specific objectives include:

- Provide precision agriculture methods for irrigation and robotized weeding operations of cereals production.
- Allow real-time monitoring of the state of the network.
- Predict the quality of the network with Machine Learning.
- Provide feedback to the operator about the quality of data to adjust current communication system capacity.

Framework: The position is part of the MSCA project **GreenFieldData** - IoRT Data management and analysis for Sustainable Agriculture, ID: 101226371. This is a double degree PhD project between **Aarhus University (Denmark)** and **Clermont Auvergne University (France)**.